E-commerce Data Analysis: Executive Brief

Objective: Understand customer purchasing behavior, product sales, payment preferences and the relationship between delivery performance and customer satisfaction.

Key findings

1. Credit cards dominate payment value.
Credit card payments accounted for approximately 78.3% of recorded payment value, making them the leading payment method. Boleto was the second-largest method.

2. Beauty and health generated the highest sales.
This category recorded 1,258,681.34 in product sales, followed by watches and gifts at 1,205,005.68.

3. Bed, table and bath led in sales volume.
The category recorded 11,115 items across 9,417 orders. Its high volume did not translate into the highest total sales value.

4. Average order value was 137.75.
The analysis covered 98,666 orders, with order values ranging from 0.85 to 13,440.00.

5. Late deliveries were associated with lower customer satisfaction.
On-time orders had an average review score of 4.29/5, compared with 2.57/5 for late orders. This 1.72-point difference highlights delivery performance as an important area for investigation.

Limitation

Some product records had no category assigned: 1,603 items across 1,451 orders. The delivery analysis also excluded orders without delivery dates or customer review scores. The results show an association between late delivery and lower ratings, not proof of causation.

Recommended next action

Investigate the causes of late deliveries, particularly by seller and delivery location. Prioritize improvements in areas with frequent delays and monitor whether customer review scores improve.

Conclusion: The analysis identifies strong-performing product categories, a clear preference for credit card payments and a substantial relationship between delivery performance and customer satisfaction. Improving delivery reliability is a practical area for further investigation.